

# Architecture & Ecosystem Design — Part 2: Shared Platform Services

## 1. Shared Services Philosophy

Shared Platform Services are the permanent foundation of the AROH ecosystem. Every service exists because it is expected to be consumed by multiple products. A service should only become part of the platform if it provides reusable business value across the ecosystem. Otherwise, it belongs inside the product that requires it.

Every shared service must satisfy the following characteristics:

- Single Source of Truth (SSOT)
- Product-agnostic
- Independently evolvable
- Versionable
- Documented
- Secure by default
- Extensible
- Observable
- Testable
- Replaceable without affecting consuming products

A shared service is never aware of the internal implementation of any product. It only exposes stable contracts.

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## 2. Identity Platform

Identity is the heart of the ecosystem.

Every user exists exactly once.

Every application references the same identity.

Identity owns:

User Registration

Authentication

Authorization

Sessions

Roles

Permissions

Profile Metadata

Email Verification

Password Management

Social Authentication (Future)

Organization Membership (Future)

Multi-device Sessions

Account Recovery

Audit Trail

Security Events

Activity History

User Preferences

Global Settings

Identity should never belong to Nebula.

Never belong to JavaPath.

Never belong to SpeDex.

Never belong to Music Mirror.

Never belong to Task Pro.

Identity belongs only to the Platform.

Future identity capabilities include:

Single Sign-On

OAuth Providers

Magic Links

Passkeys

Two-Factor Authentication

Organization Accounts

Team Accounts

Enterprise Identity

SAML

SCIM

Identity Federation

Delegated Administration

Identity Risk Detection

Adaptive Authentication

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## 3. Authentication Service

Authentication answers only one question.

Who is this user?

It never decides permissions.

Responsibilities:

Login

Logout

Signup

Refresh Tokens

Session Validation

Remember Me

Device Registration

Token Refresh

Email Verification

Password Reset

Session Expiration

Session Revocation

Multi-device Sessions

Biometric Authentication (Future)

Passkey Authentication

Authentication Provider Abstraction

Products never authenticate users themselves.

They request authentication through the platform.

Authentication providers should remain abstract.

Today:

Firebase Auth

Tomorrow:

Supabase Auth

Auth0

Clerk

Keycloak

Custom Identity

Products remain unchanged.

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## 4. Authorization Service

Authentication identifies.

Authorization decides.

Authorization owns:

Roles

Permissions

Policies

Claims

Resource Access

Product Access

Feature Access

Admin Privileges

Operator Privileges

Ownership Rules

Permission Resolution

Permission Inheritance

Future RBAC

Future ABAC

Future Policy Engine

Future Organization Permissions

Future Team Permissions

Permissions should never be hardcoded inside UI components.

Permission checks should exist within application services.

Critical operations require server authorization.

Never trust client-side permission checks.

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## 5. User Profile Service

Authentication stores identity.

Profile stores information.

Responsibilities:

Display Name

Username

Avatar

Bio

Country

Language

Timezone

Preferences

Accessibility Settings

Theme

Notification Preferences

Membership

Connected Accounts

Recent Products

Bookmarks

Achievements

Profile Visibility

AI Preferences

Search Preferences

Future Portfolio

Future Organizations

Future Teams

Future Reputation

Profile data belongs to the platform.

Every product consumes it.

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## 6. Wallet Platform

Wallet is one of the most critical services.

It should never allow client authority.

Wallet owns:

Balance

Ledger

Transactions

Credits

Debits

Rewards

Refunds

Purchases

Membership Payments

Future Marketplace Payments

Future Product Payments

Future Transfers

Future Subscriptions

Future Billing

Future Promotions

Future Coupons

Future Rewards

Future Cashback

Future Partner Credits

Wallet should never expose write access.

Products submit transaction requests.

Wallet validates.

Wallet records.

Wallet publishes events.

Products receive results.

Wallet becomes the financial authority.

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## 7. Membership Platform

Membership is not a product feature.

Membership belongs to the platform.



Responsibilities:

Plans

Subscriptions

Benefits

Entitlements

Access Rules

Upgrade Logic

Downgrade Rules

Expiration

Renewal

Grace Period

Trial Plans

Student Plans

Premium Plans

Enterprise Plans

Lifetime Plans

Partner Plans

Feature Unlocks

Membership never directly changes UI.

Products ask:

"What can this user access?"

Platform answers.

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## 8. AI Platform

AI is not another page.

AI is infrastructure.

Every product should eventually use it.

Responsibilities:

Provider Routing

Prompt Templates

Conversation Memory

Context Management

Knowledge Base

Tool Calling

Response Caching

Model Selection

Cost Tracking

Rate Limiting

Safety Filters

Logging

Analytics

Personalization

Prompt Library

Documentation Assistant

Developer Assistant

User Assistant

Support Assistant

Search Assistant

Workflow Assistant

Automation Engine

Future Multi-Agent System

Future Long-term Memory

Future AI Marketplace

Products request AI capabilities.

They never directly integrate providers.

Provider replacement becomes trivial.

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## 9. Notification Platform

Every notification originates here.

Responsibilities:

Email

Push

SMS (Future)

In-App

System Alerts

Announcements

Maintenance Notices

Security Alerts

Marketing Messages

Product Updates

Reminder Engine

Scheduling

Templates

Localization

Priority Levels

Notification Preferences

Read Status

Delivery Tracking

Retry Logic

Batch Delivery

Digest Generation

Future Cross-product Notifications

Future Workflow Notifications

Products publish events.

Notification service delivers.

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## 10. Analytics Platform

Analytics measures everything.

Responsibilities:

User Analytics

Feature Usage

Session Duration

Retention

Funnels

Performance

Business Metrics

Wallet Metrics

Membership Metrics

Product Metrics

Search Analytics

AI Analytics

Notification Analytics

Platform Health

API Metrics

Infrastructure Metrics

Error Tracking

Crash Reports

Conversion Rates

Feature Adoption

Growth Metrics

Future Predictive Analytics

Future Recommendations

Products report events.

Analytics stores.

Processes.

Visualizes.

Never duplicate analytics inside products.

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## 11. Configuration Platform

Configuration centralizes behavior.

Responsibilities:

Environment Variables

Runtime Config

Feature Flags

Maintenance Mode

Global Limits

Regional Config

AI Config

Membership Pricing

Wallet Rules

Notification Rules

Analytics Config

Theme Defaults

Security Policies

Rate Limits

API Endpoints

Platform Branding

Future A/B Testing

Future Dynamic Configuration

Configuration should never require deployments for simple operational changes.

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## 12. Product Registry

Registry knows every product.

Responsibilities:

Product Metadata

Categories

Icons

Descriptions

Routes

Status

Availability

Dependencies

Supported Platforms

Release Channels

Documentation

Licensing

Permissions

Version Compatibility

Feature Matrix

Search Metadata

Future Marketplace

Future Partner Products

Future Plugins

Homepage cards should originate from Registry.

Never hardcode products.

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## 13. Search Platform

Search belongs to the ecosystem.

Responsibilities:

Global Search

Product Search

Documentation Search

Settings Search

Help Search

Commands

Recent Searches

Recommendations

AI Search

Semantic Search

Future Cross-product Search

Future Indexed Content

Products expose searchable metadata.

Search indexes.



Returns results.

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## 14. Shared API Platform

Every public capability is exposed through APIs.

Responsibilities:

Authentication APIs

Wallet APIs

Membership APIs

Notification APIs

Analytics APIs

Registry APIs

Search APIs

Configuration APIs

AI APIs

User APIs

Health APIs

Admin APIs

Future SDK APIs

Future Public APIs

APIs expose contracts.

Never implementation.

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## 15. Shared UI Platform (ADS)

ADS owns visual consistency.

Responsibilities:

Buttons

Inputs

Forms

Cards

Tables

Dialogs

Navigation

Charts

Lists

Badges

Typography

Spacing

Icons

Colors

Responsive Grid

Accessibility

Dark Theme

Light Theme

Animations

Tokens

Every product consumes ADS.

No product creates its own design language.

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## 16. Shared Motion Platform

Motion is part of identity.

Responsibilities:

Transition Tokens

Duration Tokens

Easing Tokens

Entrance Animations

Exit Animations

Hover States

Focus States

Loading States

Scroll Animations

Reduced Motion

Gesture Standards

Micro-interactions

Future Shared Motion Library

Future Physics Engine

Every animation follows one motion language.

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## 17. SDK Platform (ASDK)

ASDK is the engineering bridge.

Responsibilities:

Schemas

Validation

Utilities

Shared Types

Hooks

Stores

API Clients

Authentication Helpers

Wallet Helpers

Configuration Helpers

Feature Flags

Permission Helpers

Error Models

Logging

Testing Utilities

Future CLI

Future Code Generators

Future Templates

Every product depends on ASDK.

Never duplicate utilities.

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## 18. Monitoring & Observability

Platform health is continuously measured.

Responsibilities:

Logging

Metrics

Tracing

Audit Logs

Health Checks

API Monitoring

Database Monitoring

Authentication Monitoring

Wallet Monitoring

AI Monitoring

Notification Monitoring

Performance Monitoring

Infrastructure Monitoring

Security Events

Incident Detection

Future Auto-Healing

Future Intelligent Alerts

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## 19. Service Communication

Products never communicate directly.

Communication always flows through platform contracts.

Product



Platform Service



Infrastructure



Platform Response



Product

This eliminates tight coupling.

Improves replaceability.

Maintains architectural consistency.

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## 20. Future Service Expansion

As AROH evolves, additional shared services may include:

Billing Platform

Marketplace Platform

Plugin Platform

Workflow Engine

Automation Platform

Content Delivery Platform

Media Platform

Localization Platform

Organization Platform

Collaboration Platform

Knowledge Platform

Recommendation Engine

Developer Portal

Public SDK Distribution

Extension Framework

Enterprise Administration

Cloud Synchronization

Offline Synchronization

Cross-device State

Voice Services

AR/VR Platform

These should be introduced only when justified by ecosystem needs. Every new shared service must reduce duplication, strengthen platform capabilities, preserve architectural consistency, and increase the long-term value of the ecosystem rather than adding unnecessary complexity.